Unit 1: Using Objects and Methods **ANSWER KEY**
Cut-And-Paste Object Creation Activity

Step 1: Cut out the code snippets that were given to you. You will arrange the code pieces to make three object classes: `Elephant`, `Giraffe`, and `Lion`. Arrange them in the correct order and glue them to a new piece of paper.

```
public class Elephant {
  private int age;
  private double weight;

  public Elephant(){
    age = 0;
    weight = 0;
  }

  public Elephant(int a, double w){
    age = a;
    weight = w;
  }

  public void loseWeight(){
    weight -= 10.5 * age;
  }

  public double calculateFoodNeeded(int days) {
     return weight * 0.05 * days;
  }

  public void talkingElephant(){
    System.out.println("The elephant's age is " + age);
    System.out.println("The elephant's weight is " + weight);
  }
}
```

```
public class Giraffe {
  private double height;
  private double neckLength;

  public Giraffe() {
    height = 4.0;
    neckLength = 0.75;
  }

  public Giraffe(double h, double n) {
    height = h;
    neckLength = n;
  }
```

```
  public void growTaller() {
    height += 0.5;
    neckLength += 0.2;
  }

  public double heightAfterYears(int years, double growthRate) {
      return height + (years * growthRate);
  }

  public void displayGiraffe() {
    System.out.println("The giraffe's height is " + height);
    System.out.println("The giraffe's neck length is " + neckLength);
  }
}

public class Lion {
  private boolean gender;
  private double roarVolume;

  public Lion() {
    gender = true;
    roarVolume = 80.0;
  }

  public Lion(boolean g, double v) {
    gender = g;
    roarVolume = v;
  }

  public void roarLouder() {
    roarVolume += 5;
  }

   public double roaringDistance() {
    return roarVolume * 1.5;
   }

  public void showLion() {
    System.out.println("The lion roars at " + roarVolume + " dB.");
  }
}
```